Group No – 46

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Hybrid Retrieval-Augmented Generation (RAG) System

1. Summary

This report documents the execution of a Hybrid Retrieval-Augmented Generation (RAG) system built using a combination of dense retrieval (Sentence Transformers + FAISS), sparse retrieval (BM25), and Reciprocal Rank Fusion (RRF). The system retrieves relevant Wikipedia documents and generates answers using a transformer-based language model (Flan-T5-base).

2. Dataset and Collection

- 200 fixed Wikipedia URLs (stored in fixed_urls.json).
- 300 randomly selected Wikipedia pages (minimum 200 words each).
- Total documents extracted: 490.
- Total chunks created after preprocessing: 1,442.

3. Preprocessing Steps

- Text cleaning (removal of extra whitespace and formatting artifacts).
- Token-based chunking: 400 tokens per chunk.
- Overlap between chunks: 50 tokens.
- Metadata stored: title, URL, chunk ID, chunk index.

4. Exploratory Data Analysis (EDA)

- Average chunks per URL: approximately 2.94.
- PCA visualization performed on dense embeddings.
- Distribution analysis of word counts and nearest neighbor distances.
- No duplicate chunk IDs detected.

```
Data columns (total 5 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   chunk_id     1442 non-null   object
 1   title        1442 non-null   object
 2   url          1442 non-null   object
 3   text         1442 non-null   object
 4   chunk_index  1442 non-null   int64
dtypes: int64(1), object(4)
memory usage: 56.5+ KB
None

### Missing Values ###
chunk_id       0
title          0
url            0
text           0
chunk_index    0
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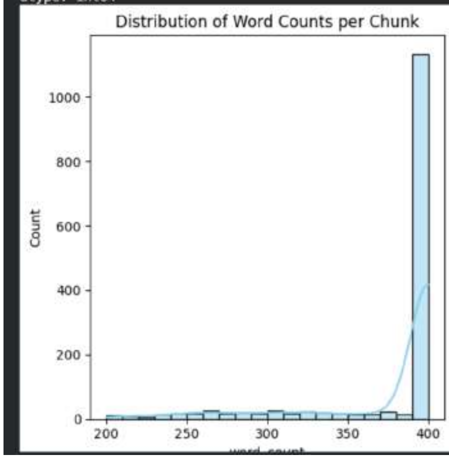
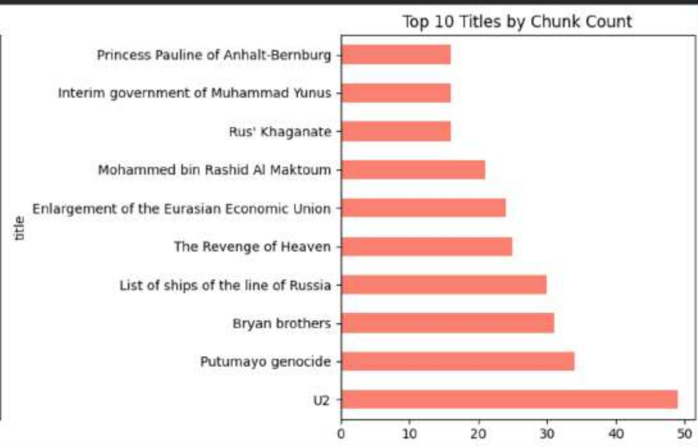
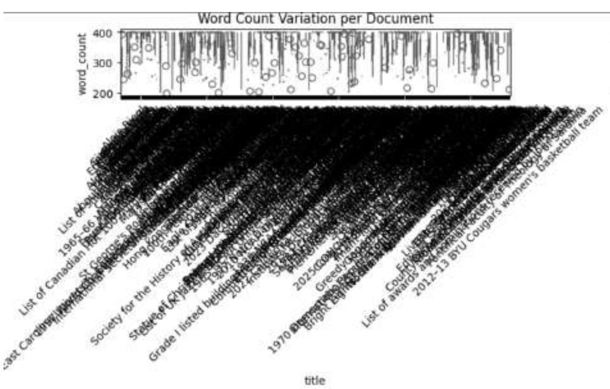
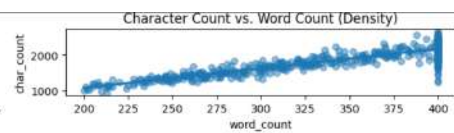
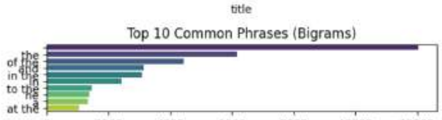
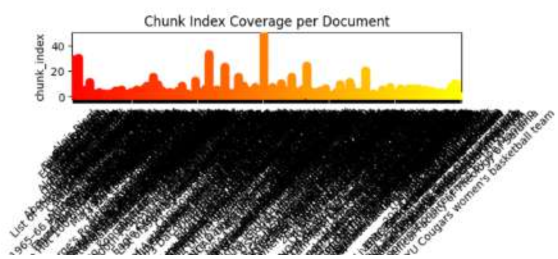




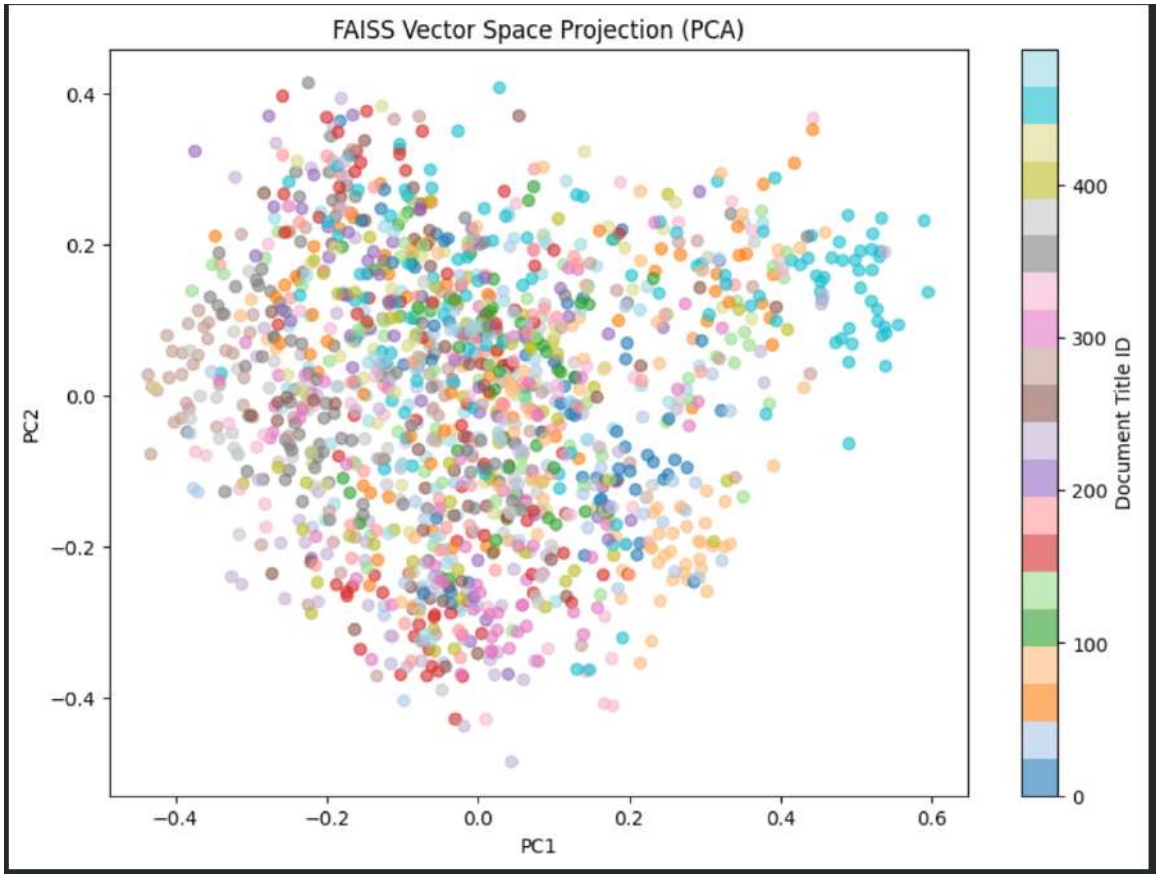


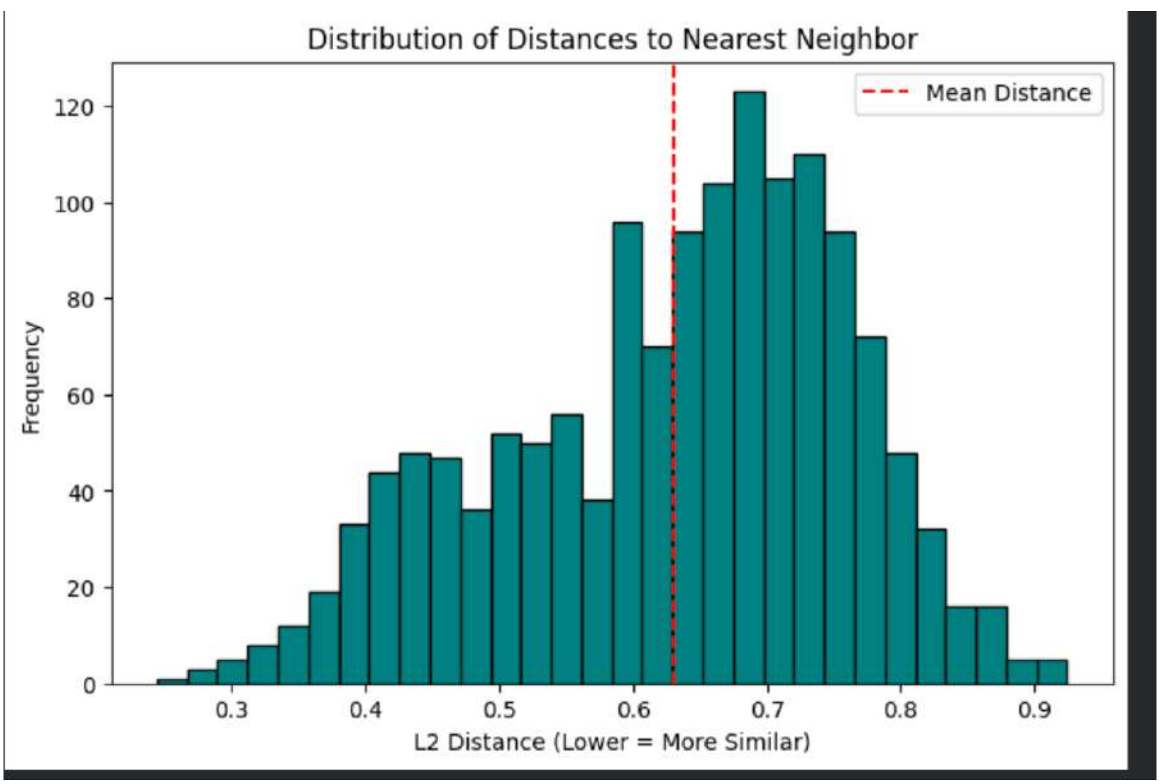




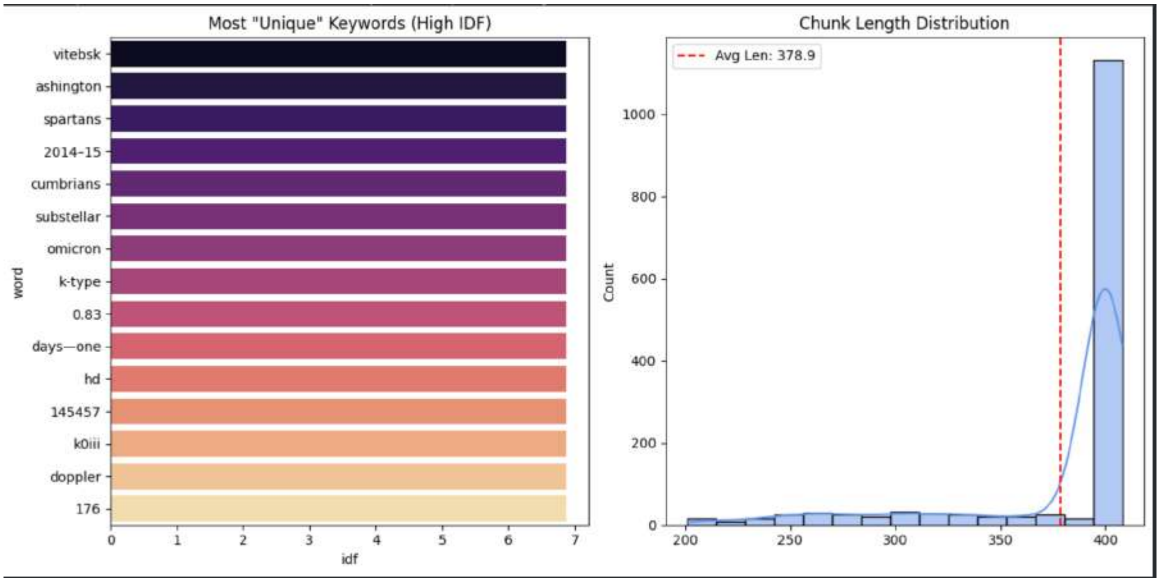


Exploratory Data Analysis (EDA) for Dense Vector





Exploratory Data Analysis (EDA) for BM25



5. Retrieval Pipeline

Dense Retrieval:

- Embedding model: all-MiniLM-L6-v2 (Sentence Transformers).
- FAISS IndexFlatIP used for similarity search.
- Embeddings normalized before indexing.

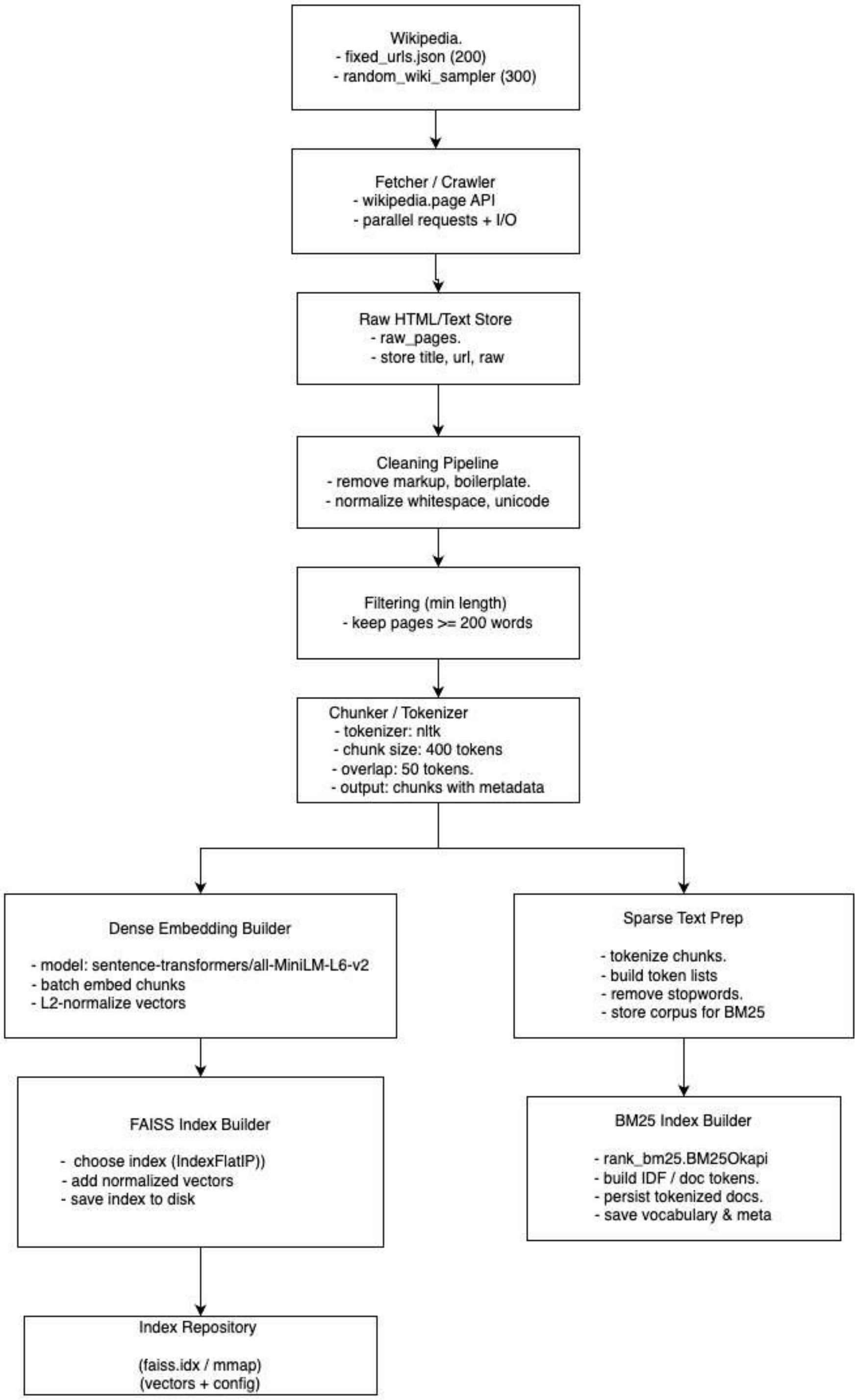
Sparse Retrieval:

- BM25 implemented using rank_bm25 library.
- IDF and document-length distributions analyzed.

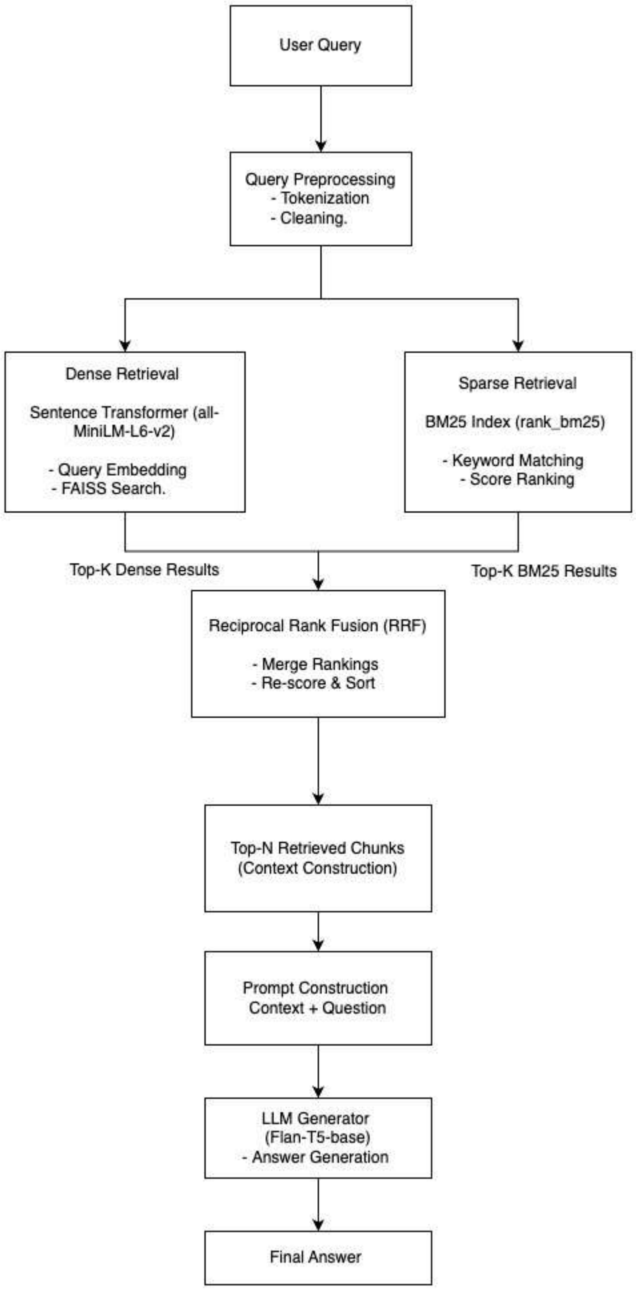
Fusion Method:

- Reciprocal Rank Fusion (RRF) combines dense and sparse rankings.

6. Indexing Pipeline Architecture



7. Retrieval Pipeline Architecture

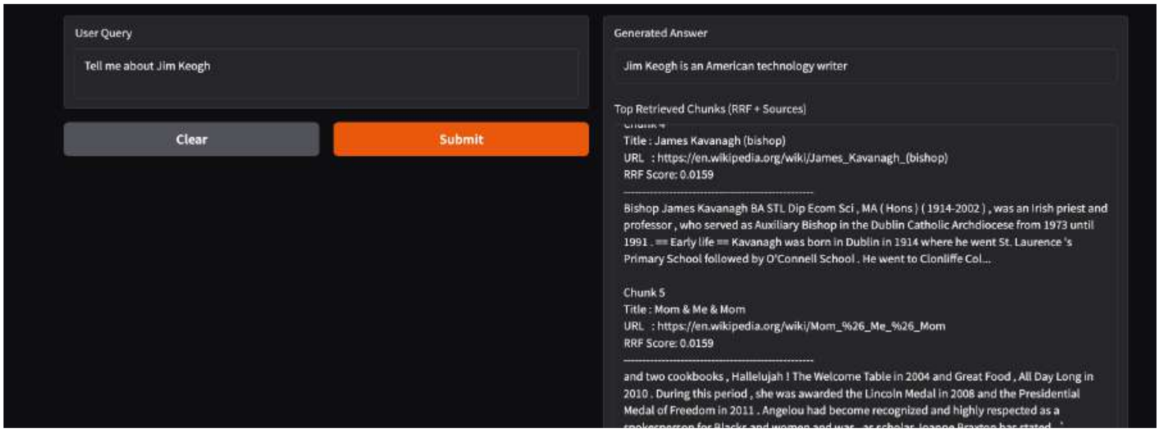


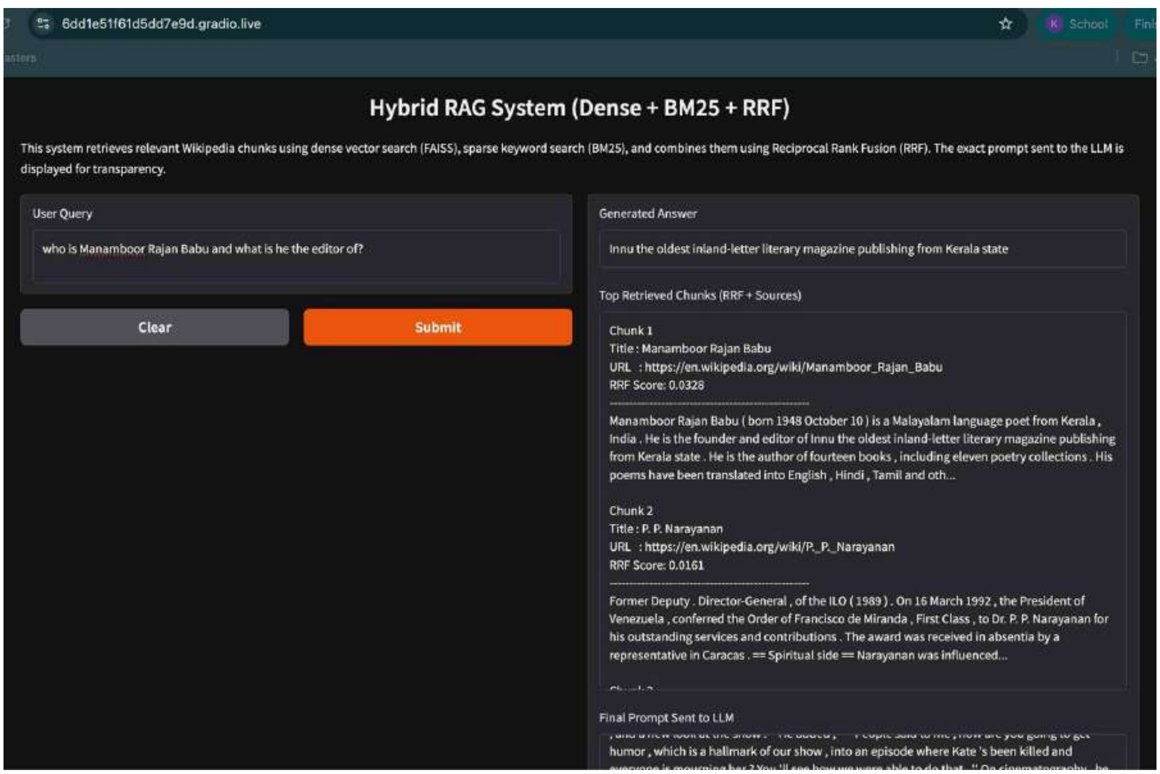
8. Generation Model

- Model used: google/flan-t5-base.
- Retrieved top-ranked chunks concatenated as context.
- Input truncated to maximum 1024 tokens.
- Final answer generated using sequence-to-sequence decoding.

9. Example Execution

- Query example: 'Tell me about Jim Keogh?'
- Retrieval results compared across Dense, BM25, and RRF.
- Example generation performed for entity-based queries.
- Response time recorded for hybrid retrieval.





10. Observations and Limitations

- Hybrid RRF improves retrieval robustness.
- Context length limited by model token constraints.

Part 2 — Automated Evaluation

Summary

This report documents the execution and results for *Part 2: Automated Evaluation* of the Hybrid RAG system. It summarizes how synthetic QA pairs were generated, the evaluation metrics used (MRR, Hit Rate, Semantic Similarity), and provides a ready-to-run evaluation script, result aggregation steps, visualization suggestions, and recommended next steps.

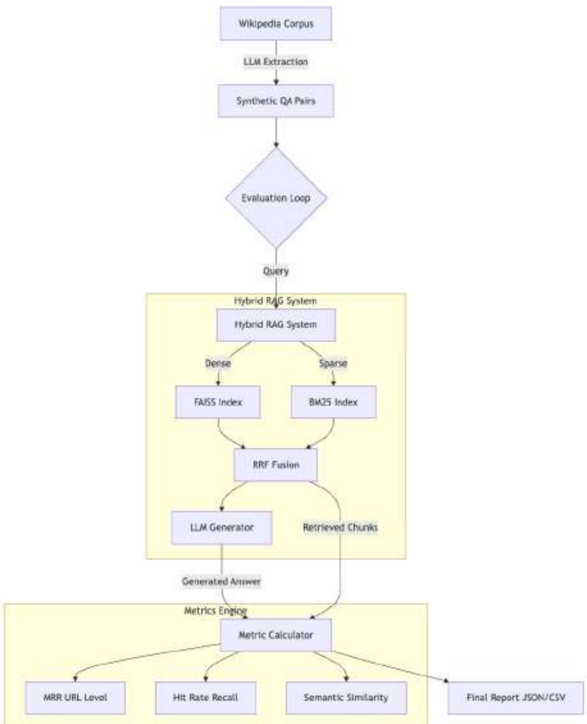
Objectives (Part 2)

- Generate a test set of synthetic questions from the corpus (100 Q&A pairs).
- Evaluate the retrieval and generation pipeline using automatic metrics:
    - Mean Reciprocal Rank (MRR) at URL level
    - Hit Rate (Recall@K)
    - Answer semantic similarity using embeddings
- Provide an automated evaluation loop and visualizations to measure end-to-end quality.

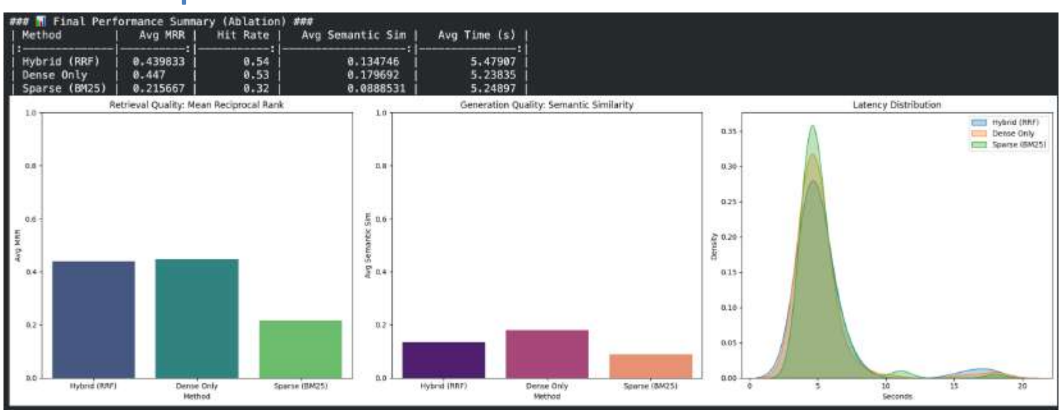
Implementation details

1. **Synthetic QA generation**
    - Randomly sample chunks from chunks (the corpus) and prompt the LLM to generate questions referring to those chunks.
    - Keep the chunk text as the ground-truth answer/context for evaluation.
2. **Evaluation metrics**
    - **MRR (URL-level):** For each question, find the first retrieved chunk whose url equals the ground-truth URL and take 1 / rank. If not found, score = 0.
    - **Hit Rate (Recall@K):** Binary 1/0 whether the ground-truth URL appears within the retrieved set.
    - **Semantic Similarity:** Encode both the generated answer and the ground-truth chunk text with all-MiniLM-L6-v2 and compute cosine similarity.
3. **Evaluation flow**
    - For each QA pair in qa_dataset (100 items):
        1. Call hybrid_rag(question, top_k=10, top_n=5) to get retrieved chunks and a generated answer.
        2. Compute MRR and Hit Rate from the rrf_chunks returned.
        3. Compute semantic similarity between the generated answer and the ground-truth chunk text.
        4. Store the per-question metrics and metadata.
    - Aggregate across all queries to compute mean MRR, overall Hit Rate, mean semantic similarity, and produce distributions/boxplots.

Execution Pipeline Architecture

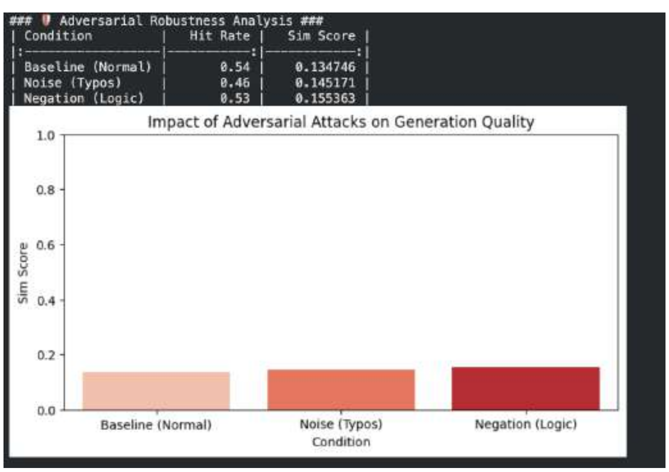


Execution Pipeline Result



Final Performance Summary (Ablation)

Method	Avg MRR	Hit Rate	Avg Semantic Sim	Avg Time (s)
Hybrid (RRF)	0.439833	0.54	0.134746	5.47907
Dense Only	0.447	0.53	0.179692	5.23835
Sparse (BM25)	0.215667	0.32	0.0888531	5.24897



Adversarial Robustness Analysis

Condition	Hit Rate	Sim Score
Baseline (Normal)	0.54	0.134746
Noise (Typos)	0.46	0.145171
Negation (Logic)	0.53	0.155363